CHEM150	**Midterm I Exam**	**Name: _____**
30 Marks	**PRACTICE A**	**Units 1-5**
8 pages	**Test time: 1.5 hours**	

Read all questions carefully.

Marks are indicated in the [ ] brackets. Each question is worth 1 mark. There are 30 questions in total.

Scientific calculators and the provided data sheets are permitted. No other aids are permitted.

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1. An element cannot _____
   A. Be part of a heterogeneous mixture.
   B. Be part of a homogeneous mixture.
   C. Be separated into other substances by chemical means.
   D. Interact with other elements to form compounds.
   E. Be a pure substance.

2. Oxygen is a _____ and nitrogen is a _____.
   A. Metal, metalloid
   B. Nonmetal, metal
   C. Metalloid, metalloid
   D. Non-metal, non-metal
   E. Non-metal, metalloid

3. Barium forms an ion with a charge of _____.
   A. +1 B. -2 C. +3 D. -3 E. +2

4. Which of the following compounds would you expect to be ionic?
   A. SF_6 B. H_2O C. H_2O_2 D. NH_3 E. CaO

5. Which metal does not form cations of differing charges?
   A. Na B. Cu C. Co D. Fe E. Sn

6. There are _____ molecules of methane in 0.123 mol of methane (CH_4).
   A. 5
   B. 2.46×10^{-2}
   C. 2.04×10^{-25}
   D. 7.40×10^{22}
   E. 0.615

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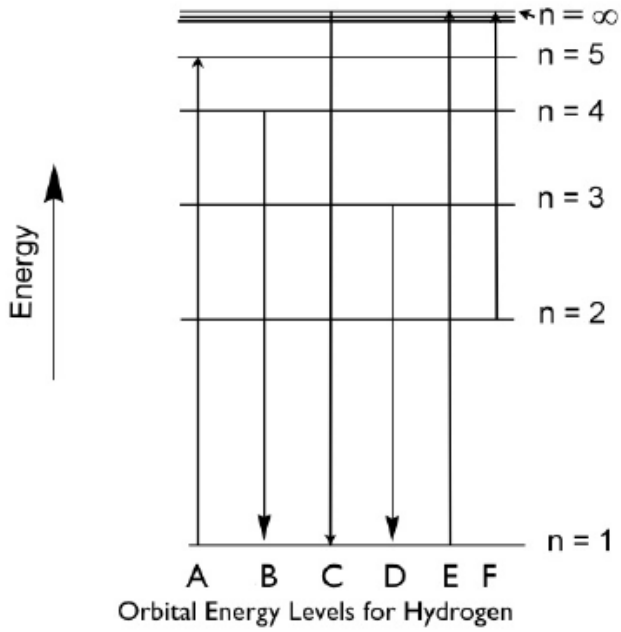
7. In 1999 scientists published the measurement of an interference pattern that demonstrated that the de Broglie wavelength of C_{60} (buckyball, mass 1.2×10^{-24} kg). Calculate the frequency (v) of the de Broglie wavelength that corresponds to a buckyball moving at a velocity of 220 m/s?

C_{60} Buckyball

   A. 1.2×10^{20} s^{-1}
   B. 2.5×10^{-12} s^{-1}
   C. 5.4×10^{17} s^{-1}
   D. 8.3×10^{-21} s^{-1}
   E. 4.0×10^{11} s^{-1}

Below is the energy level diagram for the possible energy levels of a hydrogen atom (not to scale). Answer the following questions 10-11 about the hydrogen atom.



Orbital Energy Levels for Hydrogen

8. What is the energy change (E) corresponding to the transition labeled E?
   A. -1.63×10^{-18} J
   B. 5.45×10^{-19} J
   C. 2.18×10^{-18} J
   D. 1.63×10^{-18} J
   E. -5.45×10^{-19} J

9. Consider the following chemical equation:

   $H\ (g) \rightarrow H^{+}\ (g) + e^{-}$

   Which of the transition arrows in the energy diagram best describes the energy change for this reaction?

   A. A B. B C. C D. D E. E

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10. Using the notation n, l, m_l, m_s, which of the following sets of four quantum numbers never occurs for any electron in a ground state ruthenium (Ru) atom?

A. 4, 2, -1, -1/2

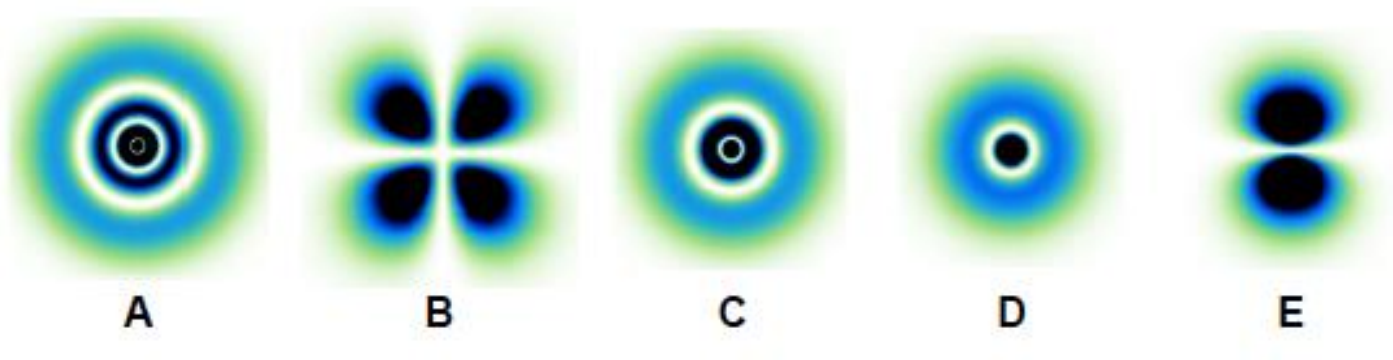
B. 5, 1, -1, +1/2

C. 2, 1, 0, -1/2

D. 5, 0, 0, -1/2

E. 3, 2, +2, -1/2

Use the following figure to answer questions 15 and 16.



11. Which of the above orbitals have no node at the nucleus?

A. A B. B C. E D. B and E E. A, C, and D

12. Which of the above orbitals have a principal quantum number of n=2?

A. A B. B C. C D. D and E E. A, C, and D

13. If A > B means the radius of A is greater than the radius of B, then which of the following comparisons is INCORRECT?

A. C > C^+ B. S > O C. As > S D. Rb^+ > Br^- E. Rb^+ > Sr^{2+}

14. If A > B means the first ionization energy (I_1) of A is greater than the first ionization energy of B, then which of the following comparisons is INCORRECT?

A. S > Si B. Be > B C. C > Si D. S > Se E. S > O

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15. Identify the atom in the following list in which the valence electron(s) experience the highest effective nuclear charge.

A. H B. He C. Li D. Be E. C

16. Which element from the set below has the largest negative (most favourable) electron affinity?

A. Mg B. Na C. Al D. Si E. P

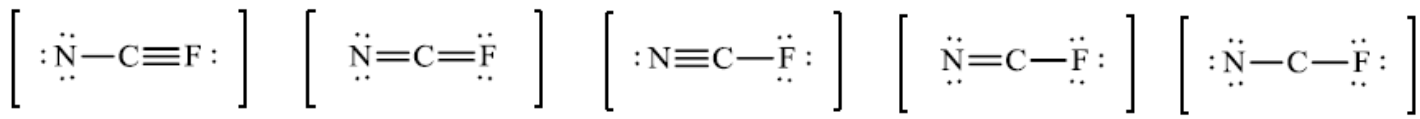
17. Which of the following pairs of ions are isoelectronic?

A. Br^- and Sr^{2+} B. Si^{4-} and C^{4-} C. O^{2-} and Cl^-

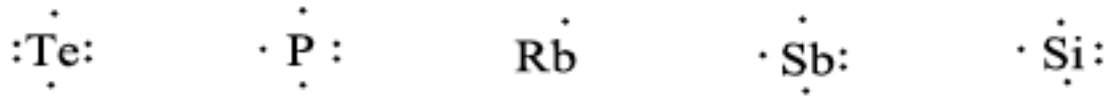
D. Si^{4-} and Sn^{4+} E. Ca^{2+} and Zn^{2+}

18. Which of these (below) is the best Lewis structure for the NCF molecule?

A B C D E



19. Which of these neutral atoms has an INCORRECT Lewis symbol?



A B C D E

20. The formal charges on the Cl atoms in the two resonance structures of $[ClO_2]^-$ are, respectively?

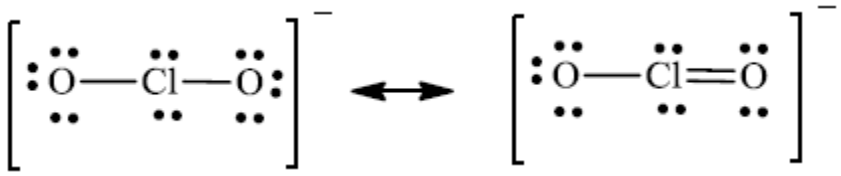
A. +1, 0

B. -1, 0

C. 0, 0

D. 0, +1

E. 0, -1



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21. How many bonds are drawn in the best Lewis structure for NO_2^+?

A. 2

B. 3

C. 4

D. 5

E. 6

22. In which one of the following molecules or ions is the octet rule not followed (not counting H)?

A. NH_3

B. NH_4^+

C. I_2

D. BH_4^-

E. NO_2

23. Of the molecules, LiF, HF, HCl, H_2O, and KCl, which compounds are covalently bonded?

A. KCl and H_2O

B. HF and KCl

C. LiF and KCl

D. LiF and HF

E. HF, HCl, and H_2O

24. Consider the Lewis structure of $HOClO_2$ in which O and Cl obey the octet rule (H is bonded to one O atom and all three O atoms are bonded to chlorine.) Which one of the following statements is INCORRECT?

A. The O atom bonded to the H atom has two nonbonded pairs of electrons.

B. The O atoms that are not bonded to the H atom have three non-bonded pairs of electrons each.

C. The O to Cl bonds are double bonds.

D. The H to O bond is a single bond.

E. Cl has one lone pair of electrons.

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25. Tungsten (W) has the highest melting point of all the pure metals (3422°C). Using your knowledge of metallic bonding, choose the best explanations for this fact from the selection below.

A. Tungsten has electrons in the 5d subshell.

B. Tungsten's molecular orbitals form a continuous band.

C. Tungsten has a half-filled s-d molecular orbital band, so the forces between atoms are of maximum strength.

D. Tungsten has as many anti-bonding electrons as bonding electrons, so the forces between atoms are of maximum strength.

E. Tungsten has a large first ionization energy, so it will not form an ionic lattice.

26. Estimate which bond angle is smallest based on the VSEPR model.

A. H-Si-H in SiH_4

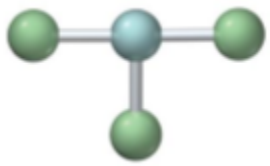
B. H-P-H in PH_3

C. H-S-H in H_2S

D. H-C-H in $H_2C{=}O$

E. H-C-H in $H_2C{=}S$

27. The molecular geometry illustrated below can only be derived from which electron domain (VSEPR) geometry?



A. Trigonal bipyramidal

B. Tetrahedral

C. V-shaped

D. See saw

E. Trigonal planar

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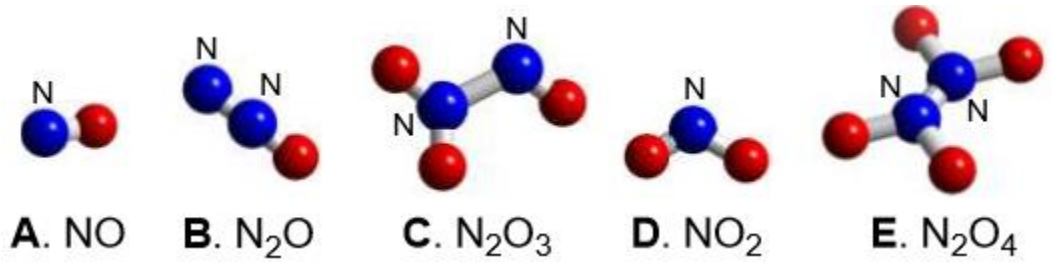
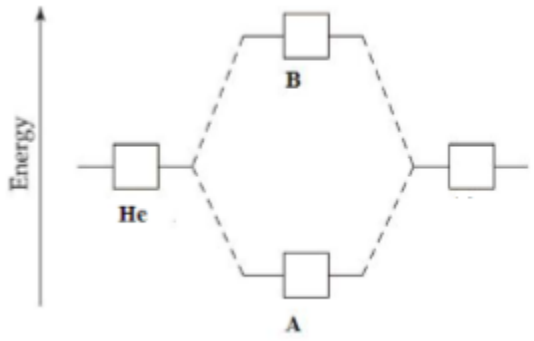
28. Which of the following oxides of nitrogen (A, B, C, D or E) is non-polar?



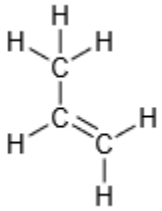
29. The following is the molecular orbital energy diagram for MOs obtained from combining the 1s orbitals of two atoms such as H or He. To correctly represent the electron energies for the ground state of dihelium +1 ion (He_2^+) the orbtials labeled A and B will contain ____ electron(s) and ____ electron(s) respectively.



A. 2, 1 B. 2, 2 C. 1, 2 D. 1, 1 E. 2, 0

30. How many sp^2 hybrid orbitals are there in a molecule of propene, C_3H_6 (structural formula shown)?

A. 2 B. 6 C. 4 D. 8 E. 16



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